

UNDERSTANDING AI

AI, Machine Learning, Deep Learning & Neural Networks

What's the Difference? — A Visual, Example-Driven Guide



AI



ML



DL



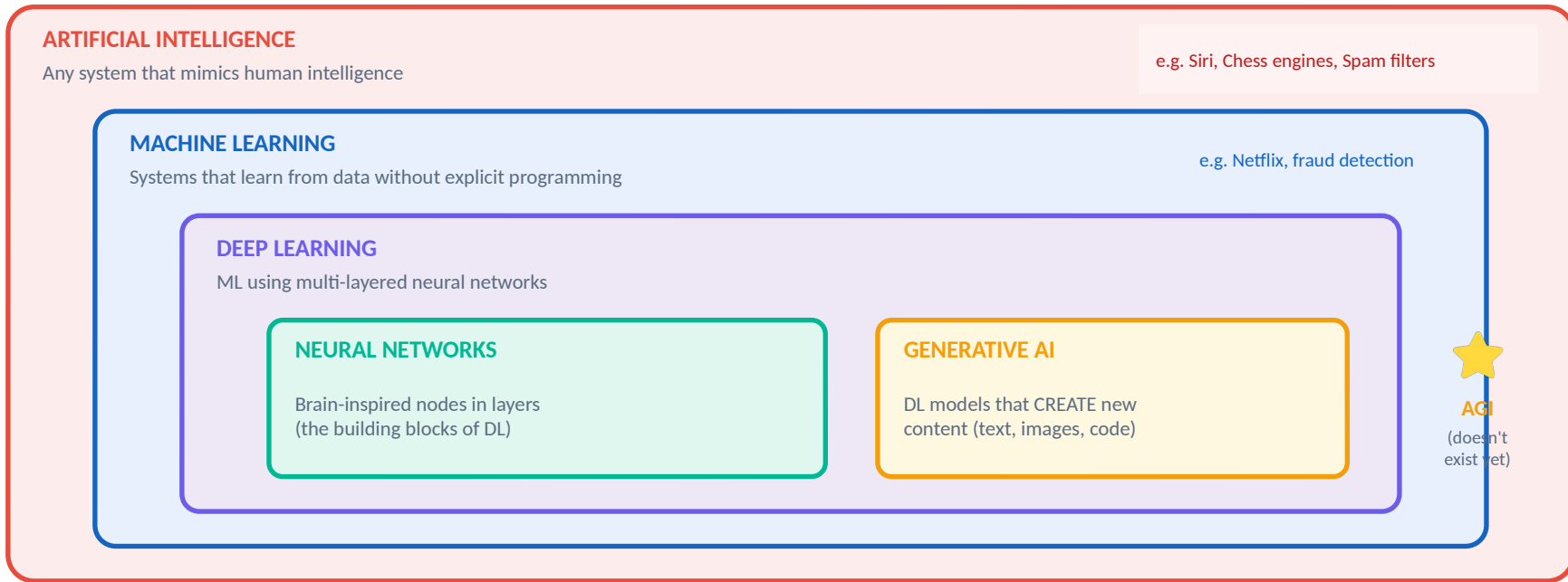
Neural
Networks



Gen AI
& AGI

Each is a subset of the previous — like Russian nesting dolls of intelligence

How They All Fit Together



Key Insight: Each layer is a SUBSET of the one outside it. All Deep Learning is ML, but not all ML is Deep Learning.

AI — The Broadest Umbrella



What is AI?

Any computer system designed to perform tasks that normally require human intelligence — reasoning, learning, perception, language understanding, decision-making.



Two Main Types

Narrow AI (ANI) — What exists today

Designed for ONE specific task. Can beat humans at that task but can't do anything else.

General AI (AGI) — Doesn't exist yet

Would match or exceed human intelligence across ALL domains. The "holy grail" — still theoretical.

AI in Your Daily Life — Both Rule-Based and Learning-Based



Parking sensors

Rule-based: IF distance < 30cm THEN beep faster



Google Search

ML-powered ranking of billions of pages by relevance



Face unlock

Deep learning recognizes YOUR face vs 7 billion others



ChatGPT / Copilot

Generative AI creates new text from learned patterns

Remember: AI is the UMBRELLA — everything else (ML, DL, NN, Gen AI) lives under it

ML — Learning From Data, Not Rules



Machine Learning: Instead of writing rules, you feed the computer DATA and let it LEARN the rules itself. The more data, the better it gets.

Traditional Programming vs. Machine Learning

Traditional (Rule-Based)

Human writes rules → Computer follows them
IF temp > 100°C AND pressure > 5bar THEN alarm



Machine Learning

Human gives data + answers → Computer learns rules
Show 10,000 engine failures → ML learns warning patterns

Three Types of Machine Learning



Supervised Learning

Labeled data: input + correct answer

Example: "These 50,000 images are cats/dogs" → Model learns to classify new images



ADAS: Labeled driving scenarios → trained object detection



Unsupervised Learning

No labels: find hidden patterns in data

Example: Customer purchase data → discover natural groupings



Warranty: Cluster failure patterns across 1M vehicles



Reinforcement Learning

Trial & error with rewards/penalties

Example: AlphaGo learned by playing millions of games



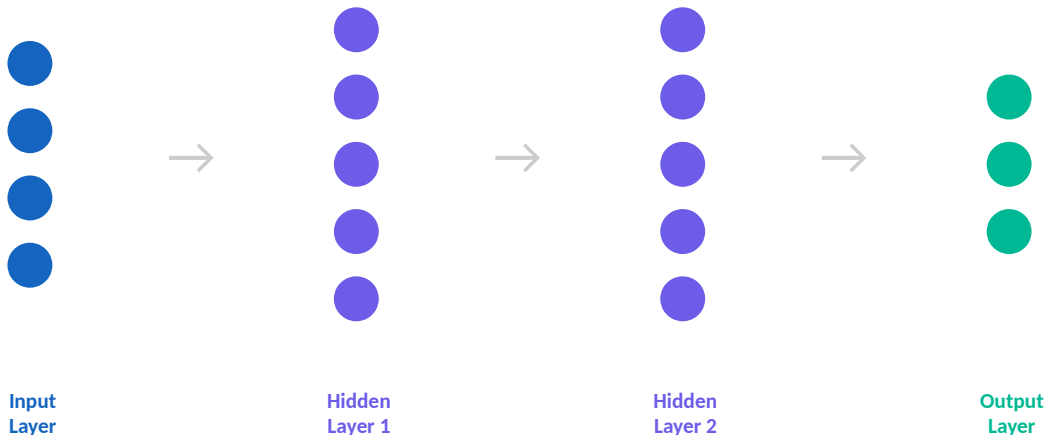
Autonomous driving: reward for safe lane changes

Neural Networks — Inspired by the Human Brain



Neural Network: A mathematical model inspired by biological neurons. Data flows through layers of connected "nodes" — each adjusting its weight until the network learns the correct output.

How a Neural Network Works



Key Concepts

Neurons (Nodes)

Mathematical functions that take inputs, apply weights, and produce an output

Weights & Biases

Adjustable parameters — training = finding the right weights through backpropagation

Activation Functions

Decide if a neuron "fires" — like a threshold. ReLU, Sigmoid, Softmax

Training = Iteration

Feed data → Check error → Adjust weights → Repeat millions of times until accurate

Automotive Analogy: A neural network is like an ECU that learns calibration maps from test data instead of being hand-tuned by engineers

Deep Learning — Neural Networks on Steroids



Deep Learning: Neural networks with MANY hidden layers (10s to 100s). "Deep" = many layers. More layers = ability to learn increasingly complex and abstract features from raw data.

Shallow vs. Deep

Shallow Neural Network (1-2 layers)

Input → 1 hidden layer → Output

Can learn simple patterns: spam detection, basic regression

Deep Neural Network (10-1000+ layers)

Input → hundreds of hidden layers → Output

Learns hierarchical features: edges → shapes → objects → scenes

Major Deep Learning Architectures



CNN

Convolutional Neural Net

Best for: Images & video



Tesla Autopilot: detects pedestrians, lanes, signs from camera feed



RNN/LSTM

Recurrent Neural Net

Best for: Sequential data



Predictive maintenance: learns time-series patterns in sensor data



Transformer

Self-Attention Model

Best for: Text, code, everything



GPT-4, Copilot, Claude — the architecture behind ALL modern LLMs



GAN

Generative Adversarial Net

Best for: Generate new data



Synthetic crash test scenarios for CAE virtual validation

Generative AI & Large Language Models



Generative AI: Deep learning models trained to CREATE new content — text, images, code, music — rather than just classify or predict. They learn the statistical structure of training data and generate similar but original outputs.

Traditional AI vs. Generative AI

Traditional AI (Analytical)

"Is this email spam?" → Yes/No

"Probability of rain?" → 73%

Classifies, predicts, detects — but never creates

Generative AI (Creative)

"Write an email about..." → Full email

"Generate a MATLAB script..." → Working code

Creates new text, code, images, audio, video

What is an LLM (Large Language Model)?



Scale

Trained on TRILLIONS of tokens from the internet — books, code, papers, forums



Architecture

Based on Transformers (2017 Google paper). Self-attention lets them understand context across long text



How They Work

Predict the NEXT TOKEN based on probability. "The car crashed into the ____" → "wall" (87%), "tree" (9%)



Key Models

GPT-4 (OpenAI), Claude (Anthropic), Gemini (Google), LLaMA (Meta) — all Transformer-based LLMs

AGI, ASI & What the Future Holds

ANI — Narrow AI

What we have TODAY

- Excels at ONE specific task
- Can't transfer knowledge across domains
- ChatGPT can write code but can't drive a car
- AlphaGo beats world champion but can't play checkers

 **Exists now**

AGI — Artificial General Intelligence

The next frontier

- Human-level intelligence across ALL tasks
- Can reason, plan, learn new skills independently
- Would pass any test a human could pass
- Estimates: 5-50 years away (highly debated)



Research stage

ASI — Artificial Superintelligence

Beyond human comprehension

- Surpasses ALL human intelligence combined
- Could solve problems we can't even formulate
- The "control problem" — how to keep it aligned
- Purely theoretical — may never happen

? Theoretical

Other Terms You'll Hear

NLP: Natural Language Processing — AI understanding human language

Computer Vision: AI that "sees" — image recognition, object detection

RAG: Retrieval-Augmented Generation — LLM + search (how Copilot works)

Fine-Tuning: Adapting a pre-trained model to a specific domain

Foundation Model: Large pre-trained model (GPT-4, Claude) adapted for many tasks

RLHF: Reinforcement Learning from Human Feedback — how ChatGPT learned to be helpful

AI vs ML vs DL vs NN — Side by Side

	AI	ML	Deep Learning	Neural Nets	Gen AI	AGI
What is it?	Umbrella term for machines mimicking intelligence	Subset of AI that learns from data	ML with many-layered neural networks	Building blocks — layers of connected nodes	DL models that create new content	AI with human-level general intelligence
Learns?	Not always (rule-based)	Yes, from data	Yes, hierarchical features	Yes, via weight adjustment	Yes, probability distributions	Would learn anything
Data needed	Varies	Moderate (thousands)	Massive (millions+)	Moderate to large	Enormous (trillions of tokens)	N/A
Example	Thermostat, chess engine	Spam filter, fraud detection	Self-driving vision, speech recognition	Image classifier, signal processing	ChatGPT, DALL-E, Copilot	Doesn't exist yet
Auto Use	ABS, cruise control logic	Warranty prediction, defect detection	ADAS object detection, NVH analysis	Sensor fusion, pattern recognition	Report drafting, code gen, docs	Fully autonomous engineering?

The Takeaway: You don't need to master the math — but understanding WHAT each term means helps you choose the right AI tool for your problem